

Prashanth Reddy Shyamala

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EDUCATION

University of Toronto, Toronto, Ontario, Canada

Sept 2022 - Apr 2027

Honours Bachelor of Science in Computer Science (ASIP Co-op Program)

GPA: 3.93 / 4.00

- Awards: UofT Excellence Award, CAD 12,000 (2026); UofT International Scholar Award, CAD 100,000 (2022).

TECHNICAL SKILLS

- **Languages:** C++, Python, TypeScript, Java, SQL
- **Machine Learning and Computer Vision:** PyTorch, CUDA, TensorRT, OpenCV, MMDetection
- **Cloud and Systems:** AWS, Terraform, Docker, PostgreSQL, React, Git, Linux

PROFESSIONAL EXPERIENCE

Software Development Engineer Co-op

May 2024 - Aug 2025

Amazon Robotics, Toronto, Ontario, Canada

- Automated continuous retraining of a production object detection model for container tracking in Amazon fulfillment centres by building an end-to-end machine learning pipeline on **AWS SageMaker**.
- Improved detection accuracy by 15 percent by implementing custom layers in **MMDetection**.
- Contributed to a workcell system that reduced container missorts by 95 percent.
- Built fault-tolerant sensor data ingestion pipelines with **AWS SQS** and **AWS Lambda**.
- Deployed a containerized, low-latency inference service on **NVIDIA Jetson** using **Docker**, **TensorRT**, **DeepStream**, and custom **CUDA** kernels, and rolled it out to real-time fulfillment systems with **AWS Greengrass**.

Undergraduate Researcher

Jan 2026 - Present

Dynamic Graphics Project, University of Toronto, Toronto, Ontario, Canada

- Developing a general framework in **Python** and **C++ (Mitsuba 3)** for unbiased inverse rendering from computational sensors, including **time-of-flight cameras**, using physically accurate volumetric techniques.
- Reduced gradient variance by combining physically based **differentiable rendering** with radiance caching via **Neural Radiance Fields (NeRF)**, keeping the estimator unbiased rather than trading correctness for speed.
- Cut memory use in **automatic differentiation** pipelines by implementing constant-memory gradient estimators in place of storing the full backpropagation graph across light paths.
- Quantified the bias introduced by approximations such as raymarching and alpha compositing, showing where they break down.
- Extended single-bounce approximations to **multi-bounce light transport** in scattering and refractive media; co-authoring a paper on the framework, currently under submission.

PROJECTS

LectureClip, Video RAG Platform (Team Capstone Project)

Jan 2026 - Apr 2026

- Built a video retrieval-augmented generation (**RAG**) platform that transcribes uploaded lectures, indexes them as embeddings, and answers natural language questions with citations to the exact segment.
- Implemented the retrieval layer over **Aurora PostgreSQL** with **pgvector**, embedding transcript text and sampled video frames into a shared vector space for cross-modal text-to-frame search.
- Built the generation layer as a **RAG** endpoint backed by **Claude** on **AWS Bedrock**, with ingestion orchestrated by **Step Functions**, **Lambda**, **Transcribe**, and **ECS Fargate**.
- Cut embedding-insert cost by 50 percent (113 ms to 56 ms per call) by profiling the transcript-to-embedding path with **cProfile** and replacing per-float **Python** string joins with a single serializer call.
- Provisioned the stack as code with **Terraform** and **GitHub Actions** CI/CD.

Rolling Shutter as IMU (Graduate Computational Imaging Project)

Oct 2025 - Present

- Investigating whether a rolling shutter sensor can act as an implicit IMU: estimated motion fields from rolling shutter distortion with a **diffusion model**, then solved for camera trajectory via **Structure from Motion (SfM)**.

EVE, Autonomous Building Assessment Robot

Jul 2024

- Won Best Environmental Hack at Hack the 6ix for an autonomous robot (**LEGO Mindstorms**, **NVIDIA Jetson**) auditing buildings against LEED and ISO 14001 standards.
- Implemented **SLAM** navigation with a **Luxonis AI camera** and image analysis with **Google Gemini**.